Rishabh Ranjan

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Personal Profile

An astrophysics student currently specialising in theoretical studies on large-scale structures and galaxy clustering. I work on building fast emulators for higher-order summary statistics obtained from large-scale structure simulations, with experience in analysing cosmological N-body simulations and inferential statistics. I am keen on exploring a diverse set of research problems in cosmology and theoretical astrophysics.

Education

2024 – Present: MSc in Astrophysics
Argelander Institut für Astronomie, University of Bonn, Germany
GPA: 1.8 on a scale of 4 (lowest) to 1 (highest)
Thesis: Evolution Mapping of the nonlinear matter bispectrum (In Progress)
Supervised by Prof. Dr. Cristiano Porciani, Dr. Alexander Eggemeier.

2019 – 2023: B.Tech (Hons) in Mechanical Engineering (with specialisation in Physics)
Indian Institute of Technology Bhilai, India
GPA: 9.09 on a scale of 4 (lowest) to 10 (highest)

Key Projects/Research

Evolution Mapping of the nonlinear matter bispectrum (Master Thesis) | 2026
LSS and Cosmology group, supervised by Prof. Dr. Cristiano Porciani and Dr. Alexander Eggemeier
- Investigated the applicability of the Evolution Mapping framework to the nonlinear matter bispectrum.
- Developed a two-staged Gaussian-process-based emulator based on the above framework to accurately emulate the nonlinear matter bispectrum for fast likelihood computations.

Effect of initial conditions on the kinematic phase timescales | 2026
Astrophysical Magnetic Fields group, supervised by Prof. Dr. Jennifer Schober
- Using numerical simulations to investigate the timescales at which the magnetic fields start growing exponentially.

Exploring the Equations of State for Dynamical Dark Energy | 2025
LSS and Cosmology group, supervised by Prof. Dr. Cristiano Porciani
- Used Bayesian techniques to constrain the equation of state parameters w_0 and w_a for different functional forms of $w_0 w_a$ dark energy models using DESI DR2 and SNe likelihoods.
- Performed model comparison across selected $w_0 w_a$ dark energy models with respect to ΛCDM.

Computational simulations of cloud disruption in an ambient medium | 2023
IIT Bhilai, supervised by Dr. Mahavir Sharma
- Simulated the phenomena of cold cloud destruction (disruption) in a hot ambient medium moving at high speeds on the PLUTO code.
- Studied the variation of the cloud disruption time with velocity of ambient medium and cloud density.

Skills

Python, C/C++, Linux (Ubuntu), Bash, Numerical modelling and simulations
Languages: English (Fluent), Hindi (Native), German (Basic)

Schools & Workshops

IHES Summer School on Cosmological Correlators
Bures-sur-Yvette, France, July 2026

Tonale Winter School on Cosmology
Tonale, Italy, December 2025

ESO-Gruber summer school: From nearby worlds to distant galaxies
European Southern Observatory, Garching, Germany, June 2025

Awards & Recognitions

Young Researchers' Award
Alumni Association IIT Bhilai, May 2023
Awarded for outstanding undergraduate research

Conference Presentations

Evolution Mapping of the Nonlinear Matter Bispectrum
Annual Meeting of the Astronomische Gesellschaft 2026
Max-Planck Institute for Astrophysics, Garching bei München, Germany, September 2026

Numerical Investigations on effect of Effect of Textured Tools on Heat Transfer during Machining
4th International Conference on Recent Advances in Mechanical Infrastructure
IITRAM Ahmedabad, India, December 2022

Impact of magnetic field on a charged fluid in kinetic theory as an application towards biofluids
International Conference on Emergent Techniques and Functional Materials
Medi-Caps University, Indore, India, July 2022

Professional Experience

Jul 2023 – Sept 2024: Model Validation Analyst
Deutsche India Private Limited (Deutsche Bank), Mumbai
- Worked on analysing model risk in financial models pertaining to the Treasury and Liquidity Management
- Developed automated validation workflows for Liquidity models.

Jan 2023 – Apr 2023: Modelling and Simulations Intern
Gamma Technologies Inc, Pune
- Developed 1D simulation models for two-wheeler electric vehicles (EVs) on GT-SUITE.

Other Projects

Effects of surface texturing on machining performance of cutting tools | 2023
IIT Bhilai, supervised by Dr. Soumya Gangopadhyay and Dr. Vijay S. Duryodhan
- Performed comparative simulations of the machining process with textured and untextured tools.
- Studied the variation of thermal properties such as tool tip temperature and heat transfer coefficient in machining using textured and untextured tools.
- Performed experimental analysis on the impact of texture geometry on performance of cutting tools.

Effect of magnetic field on viscosity of biofluids | 2022
IIT Bhilai, supervised by Dr. Sabyasachi Ghosh
- Developed a mathematical model for calculating the transport properties of biofluids in the presence of a magnetic field using Boltzmann transport equation and relaxation time approximation.
- Calculated the thermal and electrical conductivity of biofluids in presence of electric and magnetic fields.

Academic Publications

Improvement in machinability characteristics of Ti6Al4V using laser surface texturing of cutting tools with varying geometry of textures.
The International Journal of Advanced Manufacturing Technology 135, 5429–5453 (2024)
*Chandramani Upadhyay, **Rishabh Ranjan**, Anvesh Gaddam, Soumya Gangopadhyay, Vijay S. Duryodhan.*

Numerical investigations on effect of textured tools on heat transfer during machining.
AIP Conference Proceedings; 5 January 2024; 2960 (1): 040009.
*Rishabh Jain, **Rishabh Ranjan**, Vijay S. Duryodhan, Soumya Gangopadhyay*

Extracurricular Activities

Apr 2021 - Mar 2022: General Secretary of Science and Technology Affairs, IIT Bhilai
Head of the Science and Technology council at IIT Bhilai

Mar 2022: Institute Representative, Inter-IIT Tech Meet 10.0
Lead the IIT Bhilai contingent in the Inter-IIT Tech Meet 10.0

May 2020 – Apr 2021: Secretary, Space Exploration Society, IIT Bhilai
Head of the astronomy club at IIT Bhilai

Jan 2020 – Jul 2021: Executive, National Service Scheme, IIT Bhilai
Planned and conducted various social volunteering events